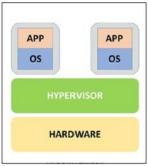

Figure 1: Type1 hypervisor

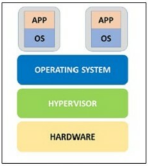

Figure 2: Type2 hypervisor

Linux kernel itself into a hypervisor and thus VMs have direct access to the hardware. KVM also contains processor-specific kernel modules such as *kvm-intel.ko* and *kvm-amd.ko*. Virt-manager and *virsh* applications are generally used to manage the virtual machines created using KVM. Virt-manager provides a GUI, whereas *virsh* is a command line utility. There is a misconception that KVM is a Type2 hypervisor that is hosted by the operating system, and not a bare metal hypervisor. This is a persistent myth, but the truth is that KVM actually does run directly on x86 hardware.

VMware free ESXi

We may think that VMware products are proprietary and not free but that's not always true. VMware's premium hypervisor product, named VMware ESXi, is available for free download[1]. Though it's not open source, some of its components source software is available for download. However, a user can choose to work for 60 days with all advanced features enabled if the free version's serial number is not entered. Try out free VMware ESXi from *https://my.vmware.com/web/vmware/evalcenter?p=free-esxi6*

Xen

The Xen Project is one of the leading open source virtualisation platforms. The Xen hypervisor is licensed under GPLv2. Like many of its competitors, Xen is also available in a commercial form from Citrix. Oracle VM is another commercial implementation of Xen. The Xen Project platform supports many cloud platforms such as Openstack, Cloudstack, etc. The Xen hypervisor provides an efficient and secure feature set for virtualisation of X86, IA64, ARM and other CPU architectures, and has been used to virtualise a wide range of guest operating systems, including Windows, Linux, Solaris and various versions of the BSD operating systems.

Microsoft Hyper-V

Microsoft has introduced Hyper-V as a competitor for many other virtualisation products. It's available for a free download[3] from an evaluation point of view. The free standalone Hyper-V server 2012 has all the features which are integrated in the Hyper-V role in Windows Server 2012 such as shared nothing live migration, failover clustering, etc. The architecture of Hyper-V is clearly explained in Wikipedia[4].

Type2 hypervisors

Xvisor

Xvisor is a Type2 monolithic open source hypervisor that aims to provide lightweight, portable and flexible virtualisation solutions. It is supported on X86 and ARM CPU architectures. One major difference is that Xvisor is completely monolithic; so it has one common software for hardware access, CPU virtualisation and guest IO emulation. However, other virtualisation technologies such as KVM and Xen are partially based on monolithic and micro-kernels, respectively. Partially monolithic hypervisors such as KVM are an extension of general-purpose monolithic OSs (such as Linux), which provide host hardware access and CPU virtualisation in the kernel and guest IO emulation from an application running in the user space (such as Qemu). Micro-kernel hypervisors are usually lightweight micro-kernels providing basic host hardware access and CPU virtualisation in the kernel, whereas the rest are dependent upon managing guests (such as Dom0 of Xen). Refer to the Xvisor website [5] for more details and to download it.